

Ashley Roselynn Vincent

arvince1@uci.edu | (650) 350-0756

EDUCATION | University of California, Irvine | Class of 2028 | B.S. Biomedical Engineering

PROJECT EXPERIENCE

UCI Rocket Project Liquids | Applications & Systems Engineer

June 2025 - Present

- Manage propulsion requirements traceability and engineering change control by consolidating design reviews and maintaining Engineering Change Notices (ECNs) to preserve design rationale, ownership, and system-level functionality as designs evolve
- Perform systems integration and verification on a liquid propulsion vehicle, evaluating requirements and compatibility across fluid, mechanical, propulsion, aerostructure, recovery, and avionics subsystems using V-model systems engineering methodology
- Lead recurring systems meetings with propulsion subsystem leads and coordinate with avionics and launch vehicle engineers to evaluate cross-system impacts of design changes
- Serve as Documentation Officer across cold-flow and vertical hot-fire test campaigns, maintaining test logs and driving requirements verification; also operate the Dewar during cold-flow testing
- Operate and calibrate a Phantom high-speed camera provided through an Amtek partnership to capture vertical test-fire footage for failure analysis
- Machine and integrate propulsion hardware, including engine shells, ablatives, nozzle components, and feed-system tubing; maintain fitting inventory
- Build sponsorship and manufacturing partnerships with Swagelok, Amtek, Allan Aircraft, SECO Seals, Thirty Two Machine, and Nikon, support \$40,000+ in sponsorships and author three technical partnership case studies

BMES @ UCI | Project Coordinator

June 2025 - Present

- Lead the MedTech Founders Program, a student innovation pipeline serving 40+ students and bridging biomedical engineering concepts with hands-on prototyping
- Design curriculum structure and cohort milestones while mentoring teams across multiple concurrent medical-device projects
- Develop and lead technical and entrepreneurial workshops covering ideation, prototyping, and pitching; organize campus-wide innovation programming with industry judges

BMES @ UCI Open MIC & UCI Engineering Design Pitch and Scramble

March 2025

- Led SeraSwaddle, a skin-to-skin-contact swaddle design (2nd place), and Inferno, a firefighting HUD integrating and visualizing operational data (3rd place)

Aragon Robotics (FRC 840 & FTC 4345) | Captain & Outreach Lead

August 2021 - June 2024

- Created IncluSTEM, an adaptive robotics curriculum making STEM workshops accessible to children with disabilities
- Established and led robotics and CAD workshops for government-school students in India, many without prior computer or internet access
- Launched EmpowerHer Showdown, a robotics competition expanding pathways for female and non-binary students into technical team roles

RESEARCH EXPERIENCE

Edwards Lifesciences Foundation CIRC | Grosberg Lab

November 2024 - Present

Undergraduate Research Assistant | Lab Manager Assistant

- Develop an in vitro ischemia-reperfusion injury model using neonatal rat ventricular myocytes on a heart-on-a-chip platform; fabricate devices, assist with cell harvests, and perform staining and experimentation
- Train custom Cellpose models for automated nuclear segmentation of live/dead staining and use a Python-based pipeline to quantify viability across experimental conditions; process contractility data using ImageJ
- Support shared lab operations by managing inventory and maintaining and sterilizing autoclaves, biosafety cabinets, incubators, and vacuum systems according to biosafety protocols

AWARDS / HONORS

- Philip S. Hiestand Engineering Research Award - \$5,000 scholarship for research excellence at UCI (June 2025)
- UROP Fall Project Pitch Finalist - research optimizing the lab's PDMS fabrication protocol
- Alpha Eta Mu Beta Initiate - national biomedical engineering honors society (December 2024)
- FIRST Engineering Inspiration Award - awarded twice at international-level competitions for expanding STEM access (March 2023 & March 2024)

SKILLS

Systems Engineering: Requirements Traceability, Verification & Validation (V&V), Systems Integration, Interface Management, Engineering Change Control, Test Documentation

Programming: C/C++ (Arduino), Python (Cellpose image analysis), MATLAB

Engineering Tools: SolidWorks, 3D Printing, Laser Cutting, Machine Shop Tools, ImageJ

Lab Techniques: Aseptic Technique, Immunostaining, PDMS Fabrication